REPORT

1. Hospital Doctor–Shift Assignment Using Backtracking Search

Problem Statement

A hospital needs to assign three doctors D1, D2, and D3 to three shifts: Morning, Afternoon, and Night. Each shift must have exactly one doctor. D1 cannot work the Night shift, D2 must work before D3, D3 cannot work the Morning shift, and all doctors must be assigned exactly one shift. Backtracking Search is used to assign the doctors step by step while checking all constraints.

Solution

Doctors: D1, D2, D3

Shifts: Morning, Afternoon, Night

Step 1: D1 → Morning

Constraint: D1 cannot work Night.

Result: Valid. Continue.

Step 2: D2 → Afternoon

D2 must work before D3. Therefore, D3 can be assigned to Night.

Step 3: D3 → Night

D3 is not in Morning and D2 is before D3. All doctors are assigned exactly once.

Final Schedule

Shift	Doctor
Morning	D1
Afternoon	D2
Night	D3

Constraint Check:

D1 ≠ Night → Satisfied

D2 before D3 → Satisfied

One doctor per shift → Satisfied

D3 ≠ Morning → Satisfied

All doctors assigned → Satisfied

Final Answer: D1 → Morning, D2 → Afternoon, D3 → Night.

2. Robot Grid Navigation Using Search

Problem Statement

A robot operates in a 5×5 grid and must move from Start (S) to Goal (G). The robot can move Up, Down, Left, or Right, with each movement costing 1. Obstacles cannot be traversed. Manhattan Distance is used as the heuristic to guide the search toward the goal and find the optimal path.

Grid

```
S . . # .
. # . # .
. # . . .
. . # # .
# . . . G
```

Solution

Manhattan Distance:

$h(n) = |x_n - xG| + |y_n - yG|$

Evaluation function:

$f(n) = g(n) + h(n)$

Step	Node Expanded	g(n)	h(n)	f(n)
1	(1,1) S	0	8	8
2	(2,1)	1	7	8
3	(3,1)	2	6	8
4	(4,1)	3	5	8
5	(4,2)	4	4	8
6	(5,2)	5	3	8

7	(5,3)	6	2	8
8	(5,4)	7	1	8
9	(5,5) G	8	0	8

Priority Queue Updates

Start: [(1,1)]

After expanding (1,1): [(2,1), (1,2)]

After expanding (2,1): [(3,1)]

After expanding (3,1): [(4,1)]

After expanding (4,1): [(4,2)]

After expanding (4,2): [(5,2)]

After expanding (5,2): [(5,3)]

After expanding (5,3): [(5,4)]

After expanding (5,4): [(5,5)]

Optimal Path

S → (2,1) → (3,1) → (4,1) → (4,2) → (5,2) → (5,3) → (5,4) → G

Number of moves = 8

Cost per move = 1

Total Cost = 8

3. Autonomous Rescue Robot Using Online Search

Problem Statement

An autonomous rescue robot operates in a disaster-affected building represented as a grid. It starts at S and must reach a survivor at G while avoiding obstacles and risky zones. The robot can move Up, Down, Left, or Right and can observe only adjacent cells. Each movement costs 1, while entering a risky zone adds an extra cost of 2. Since the robot has limited information, it must continuously update its knowledge of the environment and make safe, low-cost decisions using an online search approach.

Solution

The constraints affect the robot's decisions as follows:

Constraint	Effect on Decision
Four-direction movement	Robot can move only Up, Down, Left, or Right.
Obstacles	Blocked cells cannot be entered.
Limited sensing	Robot knows only the nearby environment.
Movement cost	Each normal movement costs 1.
Risky zones	Entering a risky zone costs 3 in total (1 + 2).
Dynamic knowledge	Robot updates its map after every observation.
Safety	Robot avoids obstacles and unnecessary risky zones.
Minimum cost	Robot selects a route with the lowest possible total cost.

Online Search Process

Start at S

↓

Observe adjacent cells

↓

Update knowledge

↓

Select safe and low-cost move

↓

Move to next cell

↓

Observe again

↓

Update knowledge

↓

Change route if required

↓

Reach survivor G

Result

The rescue robot uses online search to make decisions with incomplete information. It observes the surrounding cells, avoids obstacles, considers the additional cost of risky zones, updates its knowledge dynamically, and selects a safe path with minimum total cost.